

Dimension Reduction

Course of Statistical Learning

Gareth et al.. An Introduction to Statistical Learning, Sec 6.3

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Slides realized by Matteo Gabelli and Alberto Castellini

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- **Subset selection** and **shrinkage** control variance introducing a subset or shrinking the coefficients
- Now we explore approaches that **transform the predictors** through a technique known as **dimension reduction**

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- Let $Z_1, Z_2, \dots, Z_M$ represent M **linear combinations** of the original p predictors ($M < p$):

$$Z_m = \sum_{j=1}^p \phi_{jm} X_j$$

for some constants $\phi_{1m}, \phi_{2m}, \dots, \phi_{pm}$ and $m = 1, \dots, M$.

- We can then fit the linear regression model using least squares:

$$y_i = \theta_0 + \sum_{m=1}^M \theta_m Z_{im}, \quad i = 1, \dots, n.$$

Why Dimension reduction?

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- The dimension of the regression coefficients is **reduced**: from estimating $\beta_0, \beta_1, \dots, \beta_p$ to estimating $\theta_0, \theta_1, \dots, \theta_M$, where $M < p$.
- Dimension reduction works in **two steps**: **first**, the transformed predictors $Z_1, Z_2, \dots, Z_M$ are obtained; **second**, the model is fit using these M predictors.
- The choice of $Z_1, Z_2, \dots, Z_M$ (or equivalently, the selection of the ϕ_{jm}) can be achieved in different ways. We consider two approaches:
 - **principal components**
 - **partial least squares**

The coefficients of the reduced system θ_m

- The **reduced model** is a **special case** of the original linear regression model, in which

$$\beta_j = \sum_{m=1}^M \theta_m \phi_{jm}$$

- This is a **constraint** to the estimated β_j coefficients, introducing a potential **bias to the estimates**.
 - When p is **large** relative to n , selecting a value $M \ll p$ can significantly **reduce the variance** of the fitted coefficients.
 - When $M = p$ and all Z_m are linearly independent. **No constraints** $\Rightarrow$ No dimension reduction $\Rightarrow$ Fitting the reduced system with θ_m is **equivalent** to fit the model with coefficients β_j

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- **Principal components analysis (PCA)**: process by which principal components are computed and used to understand the data
- PCA is used to **derive a low-dimensional set of features** from a large one
- PCA is an **unsupervised approach**, since it involves only a set of features $X_1, X_2, \dots, X_p$, and no associated response Y .
- PCA is a tool used for **data visualization** but it can be useful also in **supervised learning** problems.
- We apply PCA to perform dimension reduction for **regression**: reduce the dimension of a $n \times p$ data matrix X using only **first principal component directions**

Principal Component Directions

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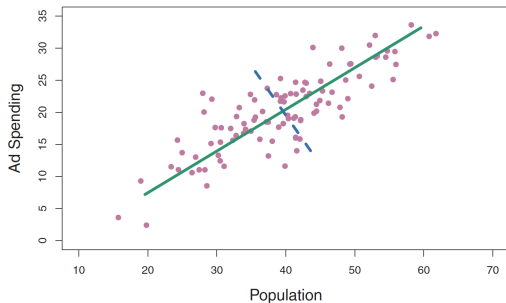
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- **Principal component directions: directions along which the original data are most variable.**
- E.g., **population size** and **ad spending** for 100 cities. **Green solid line:** *first principal component direction* of the data. **Blue dashed line:** *second principal component*

Principal Component Directions

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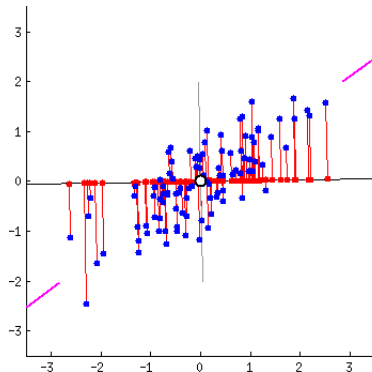
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First Principal Component

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- If we project the 100 observations onto this line, then the resulting **projected observations would have the largest possible variance**
- Projecting the observations onto any **other line** would yield projected observations with **lower variance**

- How can it be summarized mathematically? It is given by the formula

$$Z_1 = 0.839 \times (pop - \overline{pop}) + 0.544 \times (ad - \overline{ad})$$

- Here, $\phi_{11} = 0.839$ and $\phi_{21} = 0.544$ are the principal component **loadings**, which define the **direction**. $\overline{ad}$ and $\overline{pop}$ indicate the **sample mean**

The linear combination maximizing the variance.

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- **Idea:** out of every possible linear combination of pop and ad such that $\phi_{11}^2 + \phi_{21}^2 = 1$, this particular linear combination yields the **highest variance**. Namely,

$$Var (\phi_{11} \times (pop - \overline{pop}) + \phi_{21} \times (ad - \overline{ad}))$$

is maximized.

- Since $n = 100$, pop and ad are vectors of length 100. The elements of vector Z_1 are

$$z_{i1} = 0.839 \times (pop_i - \overline{pop}) + 0.544 \times (ad_i - \overline{ad}).$$

- The values of $z_{11}, \dots, z_{n1}$ are known as the **principal component scores**

Geometrical interpretation

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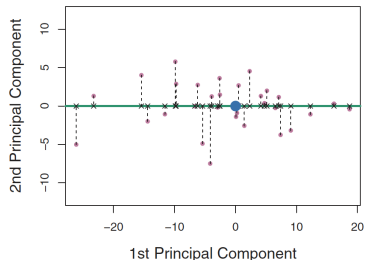
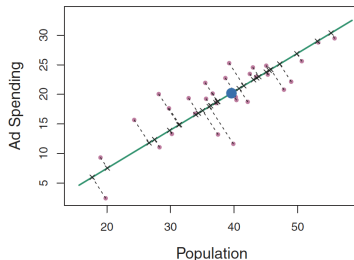
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- **Loading vector:** defines the line that is **as close as possible to the data**
- The first principal component line **minimizes the sum of the squared perpendicular distances between each point**. The line has been chosen so that the projected observations are as close as possible to the original observations.
- The corresponding **loadings** are $\phi_{11} = 0.839$ and $\phi_{21} = 0.544$

More generally...

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- Consider n observations with measurements on a set of p features, $X_1, X_2, \dots, X_p$. **PCA finds a low-dimensional representation of a data set that contains as much as possible of the variation.**
- **Main idea:** each of the n observations lives in a p -dimensional space. **Not all of these dimensions are equally interesting**
- PCA seeks a **small number of dimensions** that are as interesting as possible, where the concept of interest is measured by the amount that the observations **vary** along each dimension
- Each dimension found by PCA is a **linear combination** of the p features.
- How can we find these dimensions?

The principal component loading vector

- The **first principal component** of a set of p features is the normalized (i.e., $\sum_{j=1}^p \phi_{j1}^2 = 1$) linear combination of the features

$$Z_1 = \phi_{11}X_1 + \phi_{21}X_2 + \dots + \phi_{p1}X_p$$

with the **largest variance**

- Elements $\phi_{11}, \dots, \phi_{p1}$ are the *loadings* of the first principal component
- We solve the follow **optimization problem** over the components of the loading vector

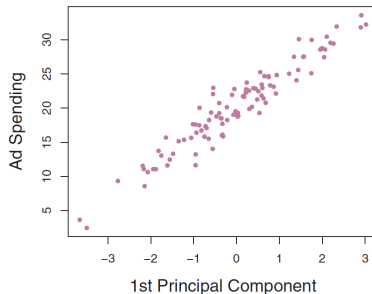
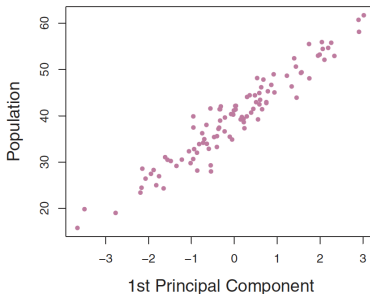
$$\max_{\phi_{11}, \dots, \phi_{p1}} \left\{ \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \phi_{j1} x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^p \phi_{j1}^2 = 1$$

How well can Z_1 represent both *pop* and *ad*?

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Since *pop* and *ad* have approximately a **linear relationship**, we might expect that a single-number summary will work well.



The plots show a **strong relationship** between the **first principal component** and the **two features** (*pop* and *ad*) that capture most of the information.

Second Principal Component

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- We can construct up to p distinct principal components.
- The **second principal component** Z_2 is a linear combination of the variables that is **uncorrelated** with Z_1 , and has **largest variance** subject to this constraint
- The **zero correlation condition** of Z_1 with Z_2 means the two components must be **perpendicular**, or **orthogonal**
- The **second principal component** is given by the formula

$$Z_2 = 0.544 \times (pop - \overline{pop}) + 0.839 \times (ad - \overline{ad})$$

- Since $p = 2$, the two principal components contain all of the information in pop and ad

First vs Second Principal Direction

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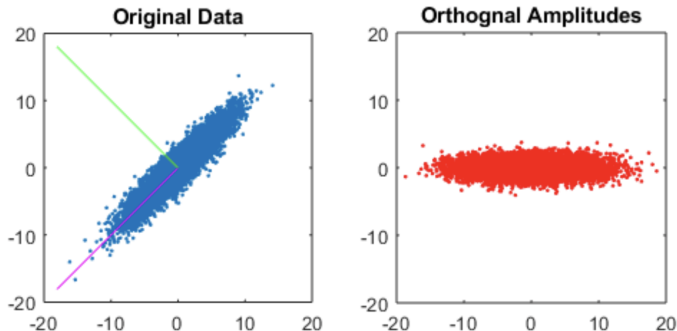
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Consider the larger variability of z_{i1} (the x-axis) versus z_{i2} (the y-axis): **the second principal component scores** are much **closer to zero** indicates that this component captures **less information**

Second Principal Component

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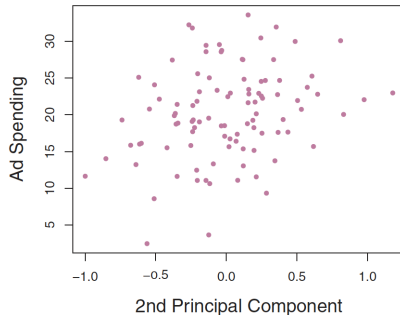
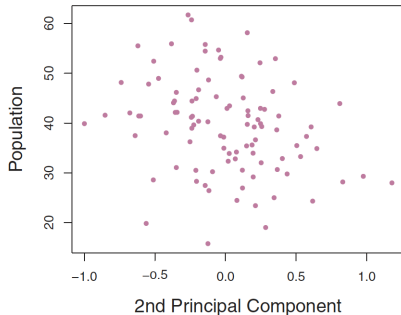
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Plots of the **second principal component scores** z_{i2} versus *pop* and *ad*: the **relationships are weak** between the second principal component and the two predictors $\Rightarrow$ Only the **first** principal component is **needed** to accurately represent *pop* and *ad*

- If we had **other predictors**, such as population age, income level, education, and so forth, then **additional components could be constructed**.
- They would **successively maximize the variance**, subject to the **constraint** of being **uncorrelated** with the preceding components.

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The Principal Components Regression Approach

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- **Principal components regression (PCR)** constructs the first M principal components, $Z_1, \dots, Z_M$, and then uses them as predictors in a **linear regression model** fit by **least squares**
- **Key idea:** often a **small number of principal components suffice** to explain most of the variability in the data, as well as the relationship with the response
- **Assumption:** the directions showing the most variation are the most **associated with Y**
- This assumption is **not guaranteed to be true** but it is often reasonable and provides good results

Advantages from PCR

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- If the previous **assumption** holds then the PCR is a valid technique
- Fitting a least squares model to $Z_1, \dots, Z_M$ will lead to **better results** than fitting a least squares model to $X_1, \dots, X_p$, since most or all of the information in the data that relates to the response is contained in $Z_1, \dots, Z_M$
- By estimating only $M \ll p$ coefficients we can **mitigate over-fitting**.
- E.g., **advertising data**: the first principal component explains most of the variance in both *pop* and *ad*. A principal component regression that uses this single variable to predict *sales* performs well.

PCR with simulated data set

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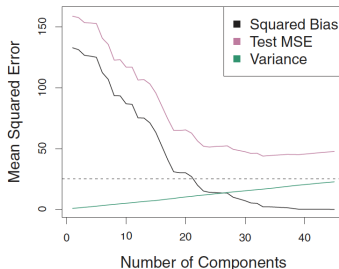
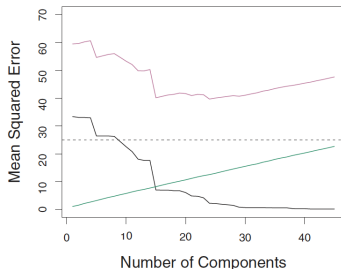
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- **PCR fits on simulated data sets ($n = 50$ and $p = 45$).** The response in the **first dataset** was a function of all the predictors. The response in the **second dataset** was generated using only two predictors.
- Performing PCR with an **appropriate choice of M** can result in a substantial **improvement** over least squares. PCR performs well when the **first few principal components** are **sufficient** to capture most of the **variation** in the predictors.

PCR versus Ridge Regression

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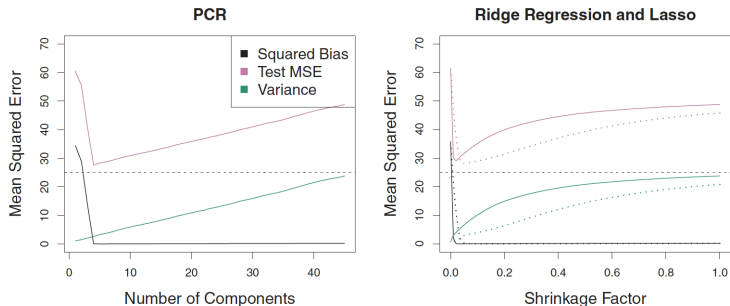
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- **First plot:** results from another simulated data set designed to be more favorable to PCR (it depends exclusively on the first five principal components). The bias drops to zero rapidly as M increases. The mean squared error displays a **clear minimum** at $M = 5$
- **Second plot:** same data. PCR (left) and ridge (dotted right) outperform lasso regression (solid, right)

PCR does not perform feature selection

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- Even though PCR provides a simple way to perform regression using $M < p$ predictors, **it is not a feature selection method**.
- Each of the M **principal components** used in the regression is a linear combination of all p of **all original features**.
- For instance, the first principal component Z_1 was a **linear combination of both pop and ad**.
- In this sense **PCR is more closely related to ridge regression than to the lasso**.
- **Ridge regression** can be considered as a **continuous version of PCR**.

CV to calibrate M

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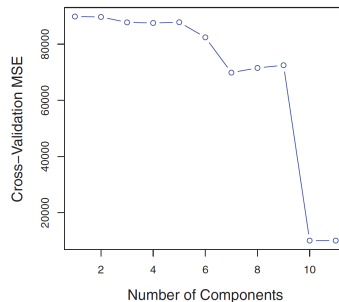
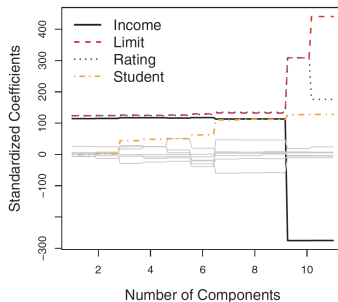
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- In PCR, the **number of principal components, M** , is typically chosen by **cross-validation**.
- In the example the lowest cross-validation error occurs when there are $M = 10$ components (no dimension reduction is applied).

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Partial Least Squares (PLS)

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- **PCR**: involves identifying **directions that best represent the predictors** $X_1, \dots, X_p$. These directions are identified in an **unsupervised way**, since the response Y is not used to determine the principal component directions. That is, **the response does not supervise the identification of the principal components**
Main drawback: no guarantee that the directions that best **explain the predictors** will also be the best directions to use for **predicting the response**
- **PLS: supervised alternative to PCR**. PLS is a dimension reduction method, which first identifies a new set of **features** $Z_1, \dots, Z_M$ that are linear combinations of the original features, and then fits a linear model via least squares using these M new features. But unlike PCR, **PLS identifies these new features in a supervised way**.

PLS coefficients are directly related to the response

- PLS makes **use of the response Y** to **identify new features** that not only approximate the old features well, but also that are **related to the response**
- **Idea:** find directions that help explain both the response and the predictors
- After **standardizing** the p predictors, PLS computes the **first direction Z_1** by setting each ϕ_{j1} equal to the **coefficient from the simple linear regression of Y onto X_j** : this coefficient is proportional to the **correlation between Y and X_j** . Hence, in computing

$$Z_1 = \sum_{j=1}^p \phi_{j1} X_j ,$$

PLS places the **highest weight** on the **variables** that are most strongly **related to the response**.

First PLS direction

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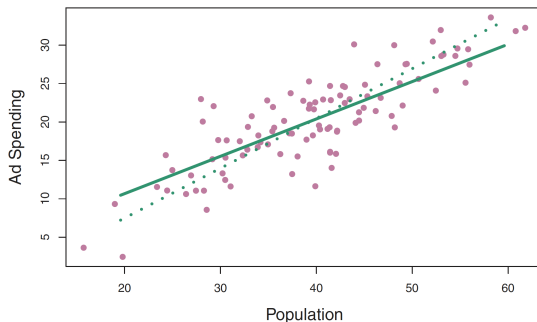
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PLS on the advertising data: the **green line** indicates the first PLS direction, while the **dotted line** shows the first principal component direction. **PLS** has chosen a direction that has less change in the *ad* dimension per unit change in the *pop* dimension, relative to PCA. This suggests that **pop is more highly correlated with the response than is ad**

PLS directions

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- The **PLS direction** does **not fit the predictors** (as closely as does PCA), but **it better explains the response**.
- **Second PLS direction**: we first **adjust** each of the **variables** for Z_1 , by regressing each variable on Z_1 and taking the **residuals** (remaining information not explained by the first PLS direction). We then **compute** Z_2 using this orthogonalized data in the **same fashion as** Z_1 was computed based on the original data. **This iterative approach** can be repeated **M times** to identify PLS components $Z_1, \dots, Z_M$.
- At the end of this procedure, we use **least squares** to fit a linear model to **predict Y using $Z_1, \dots, Z_M$** .
- As with PCR, the **number M of PLS directions** is a **tuning parameter** typically chosen by **cross-validation**.

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Low-Dimensional Data

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Most **traditional statistical techniques for regression and classification** are intended for the **low-dimensional** setting in which n , the number of observations, is much greater than p , the number of features.

For instance, consider the task of developing a model to predict a **patient's blood pressure on the basis of his or her age, gender, and body mass index (BMI)**. There are **three predictors**, or four if an intercept is included in the model, and perhaps several **thousand patients** for whom blood pressure and age, gender, and BMI are available. Hence $n \geq p$, and so the problem is **low-dimensional**. (By dimension here we are referring to the size of p .)

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In the past 20 years, **new technologies** have changed the way that data are collected. It is now commonplace to collect an almost unlimited number of feature measurements (p very large). While p **can be extremely large**, the number of observations n is often limited (due to cost or sample availability). Two **examples** are as follows:

- In order to predict **blood pressure**, one might also include measurements for **half a million single nucleotide polymorphisms (SNPs)**, i.e., individual DNA mutations, in the predictive model. Then $n \approx 200$ and $p \approx 500000$
- A **marketing analyst** interested in understanding people's **online shopping patterns** could treat as features all of the **search terms entered by users of a search engine** (known as the **"bag-of-words"** model). For a given user, each of the p search terms is scored present (0) or absent (1), creating a large binary feature vector.

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Data sets containing **more features than observations** are often referred to as **high-dimensional**. Classical approaches such as **least squares linear regression are not appropriate** in this setting.

The principal issues, that arise in the analysis of high-dimensional, are the role of the **bias-variance trade-off** and the danger of **overfitting**, that can become particularly important when the number of features is very large with respect to the number of observations.

These considerations also apply if p is **slightly smaller** than n , when we are performing supervised learning.

What Goes Wrong in High Dimensions?

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We begin by examining what can go wrong if we apply a statistical technique not intended for the high-dimensional setting. For this purpose, we examine **least squares regression**, but the same concepts apply to logistic regression, linear discriminant analysis and other classical statistical approaches.

When the **number of features p is as large as, or larger than, the number of observations n** , **least squares cannot (or rather, should not) be performed**. The reason is simple: regardless of whether or not there truly is a relationship between the features and the response, **least squares will yield a set of coefficient estimates that result in a perfect fit to the data**, such that the residuals are zero.

$p = 1$ with $n = 20$ and $n = 2$

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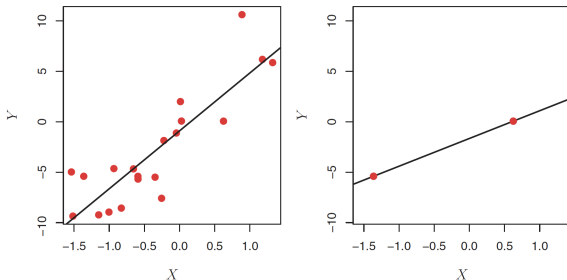
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When there are **20 observations**, $n > p$ and the least squares regression line does not perfectly fit the data. When there are only **2 observations**, then the **regression line will fit the data exactly**. This is problematic because this perfect fit will almost certainly lead to **overfitting** of the data: the resulting linear model will **perform extremely poorly on an independent test set**: it does not constitute a useful model, since is too flexible.

Training set MSE versus Test set MSE

Dimension
Reduction

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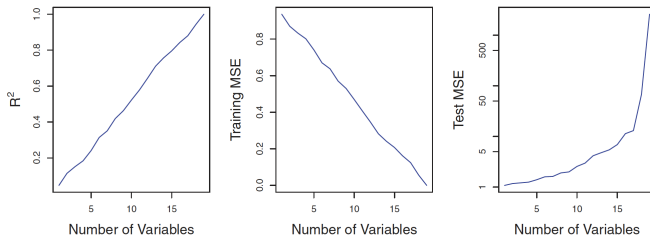
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Data were **simulated with $n = 20$** and regression was performed with p **between 1 and 20** with **each feature completely unrelated to the response**. The model R^2 **increases to 1 as the number of features** included in the model **increases**, and the corresponding *training set MSE* decreases to 0 as the number of features increases, even though the **features are completely unrelated to the response**. Conversely, the MSE on an independent **test set** becomes extremely **large** as the number of features included in the model increases.

Model performance on independent test

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Including the **additional predictors** leads to a **huge increase in the variance of the coefficient estimates**. Looking at the *test set MSE*, it is clear that the best model contains at most a few variables. However, someone who carelessly examines only the R^2 or the training set *MSE* might **erroneously** conclude that the model with the greatest number of variables is best. This indicates the **importance of applying extra care when analyzing data sets with a large number of variables**, and of **always evaluating model performance on an independent test set**.

We saw a **number of approaches** for adjusting the training set *RSS* or R^2 in order **to account for the number of variables** used to fit a least squares model. Unfortunately, the C_p , *AIC*, and *BIC* approaches are **not appropriate in the high-dimensional setting**, because **estimating $\hat{\sigma}^2$ is problematic**.

Regression in High Dimensions

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It turns out that many of the methods for fitting **less flexible** least squares models, such as **forward stepwise selection**, **ridge regression**, **the lasso**, and **principal components regression**, are particularly **useful** for performing regression in the **high-dimensional setting**.

Essentially, these approaches **avoid overfitting** by using a **less flexible fitting approach** than least squares.

As an **example**, we consider the performance of the **lasso** in a simple simulated example. There are $p = 20, 50$ or 2000 features, of which 20 **are truly associated with the outcome**. The lasso was performed on $n = 100$ training observations, and the **mean squared error was evaluated on an independent test set**.

Lasso in High Dimensions

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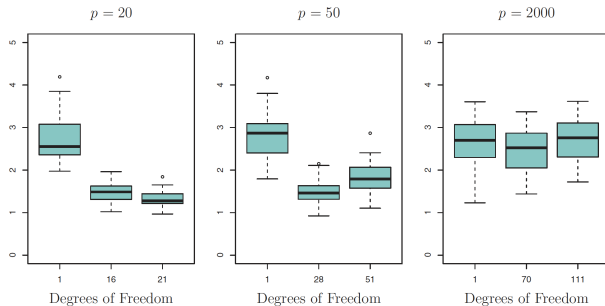
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As the number of features increases, the test error increases. In each box-plot, we use the **degrees of freedom** (the number of non-zero coefficient estimates in the lasso solution) to measure the **flexibility** of the fit. When $p = 20$, the lowest validation set error was achieved when $\lambda = 0$. However, when p **was larger** then the lowest validation set error was achieved using a **larger value of λ** .

Lasso in High Dimensions

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The plot highlights three important points:

- 1 **regularization** or shrinkage plays a **key role in high-dimensional problems**;
- 2 appropriate **tuning parameter selection is crucial** for good predictive performance;
- 3 the **test error** tends to **increase** as the **dimensionality of the problem** (i.e. the number of features or predictors) increases, unless the additional features are truly associated with the response.

Curse of Dimensionality

Dimension
Reduction

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The **third point** is in fact a key principle in the analysis of high-dimensional data, which is known as the **curse of dimensionality**. One might think that **as the number of features** used to fit a model **increases**, the **quality** of the fitted model will increase as well, but this is **not** necessarily true.

In general, **adding additional signal features** that are **truly associated** with the response will **improve** the fitted model, in the sense of leading to a **reduction in test set error**.

However, adding **noisy features** that are not truly associated with the response will lead to a **deterioration in the fitted model**, and consequently an **increased test set error**. This is because noisy features **increase the dimensionality of the problem and the risk of overfitting**.